

Semantic Search Evaluation

語意搜尋評估

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ABSTRACT | 摘要

We propose a novel method for evaluating the performance of a content search system that measures the semantic match between a query and the results returned by the search system. We introduce a metric called “on-topic rate” to measure the percentage of results that are relevant to the query. To achieve this, we design a pipeline that defines a golden query set, retrieves the top K results for each query, and sends calls to GPT 3.5 with formulated prompts. Our semantic evaluation pipeline helps identify common failure patterns and goals against the metric for relevance improvements.

► **中文譯文** 我們提出一種評估內容搜尋系統表現的新方法，用來衡量查詢與系統回傳結果之間的語意匹配程度。我們提出一項名為「切題率（on-topic rate）」的指標，用來衡量結果中有多少比例與查詢相關。為達成此目的，我們設計了一套管線：定義黃金查詢集、為每個查詢取回前 K 筆結果，並以設計好的提示詞呼叫 GPT-3.5。我們的語意評估管線有助於找出常見的失敗型態，並依此指標設定相關性改善的目標。

KEYWORDS: Semantic Relevance, Search System Evaluation, Content Search, Information Retrieval, Generative AI

► **中文譯文** 關鍵詞：語意相關性、搜尋系統評估、內容搜尋、資訊檢索、生成式 AI

1 INTRODUCTION | 前言

LinkedIn search has made significant progress over the years by incorporating semantic matching capabilities. Semantic matching capability helps members find knowledge more easily by delivering results that are conceptually related to their search query, even if they do not contain the exact keywords used in the query. As we continue to improve our system, we strive to overcome the complex quality challenges that arise:

► **中文譯文** LinkedIn 搜尋多年來透過導入語意匹配能力，已有顯著進展。語意匹配能力能讓會員更容易找到知識：即使結果沒有包含查詢裡使用的確切關鍵字，只要在概念上與查詢相關，系統依然能夠回傳。在持續改進系統的過程中，我們致力於克服以下複雜的品質挑戰：

- Indirect measurement: While we have existing engagement metrics from online experiments, they do not necessarily capture the quality of our search results. When member feedback comes

in, there is an added overhead in determining if the system is operating as expected.

- Not operationalized: Search patterns and expectations change over time, so we need an automated way to continuously measure quality.

► **中文譯文** . 間接量測：雖然我們已有來自線上實驗的互動指標，但這些指標未必能真正反映搜尋結果的品質。當會員回饋出現時，還要額外花力氣判斷系統是否如預期運作。

. 未被操作化：搜尋型態與期望會隨時間改變，因此我們需要一套自動化的方法來持續衡量品質。

To address these gaps, we present a semantic evaluation pipeline that leverages Generative AI (GAI) to evaluate quality. Our main contributions and business impacts are:

► **中文譯文** 為了補上這些缺口，我們提出一套借助生成式 AI (GAI) 來評估品質的語意評估管線。我們的主要貢獻與商業影響如下：

- We propose the metric “on-topic rate” to measure the percentage of content search results that are relevant to the intended topic. This metric serves as a tool for evaluating the performance of the content search model, and can also help identify common failure patterns. By setting goals based on this metric, we can work towards improving the relevance of search results.

► **中文譯文** . 我們提出「切題率」指標，用來衡量內容搜尋結果中，有多少比例確實與使用者意圖主題相關。此指標可作為評估內容搜尋模型表現的工具，也有助於找出常見的失敗型態。以此指標設定目標，我們就能朝改善搜尋結果相關性的方向努力。

- We present a novel semantic evaluation pipeline for search engine offline evaluation. This framework helps translate user problems into technical patterns that can be operationalized and plays a critical role for search engine offline evaluation.

► **中文譯文** . 我們提出一套用於搜尋引擎離線評估的新語意評估管線。此框架有助於把使用者的問題轉譯成可操作化的技術型態，在搜尋引擎的離線評估中扮演關鍵角色。

- We present two approaches to evaluate the Generative AI evaluation output, including conducting human evaluation and preparing a validation set to ensure the quality.

► **中文譯文** . 我們提出兩種評估生成式 AI 輸出結果的做法，包括進行人工評估，以及建立驗證集以確保品質。

2 RELATED WORK | 相關研究

Assessing the quality of search engine results has become a challenging task, particularly in determining which ranking models perform best under specific business metrics [2, 8]. The most reliable method for evaluating model performance is through online A/B testing, but this approach has limitations [5, 7]. Firstly, due to the high cost of traffic, only a limited number of models can be compared within a given timeframe, as it requires a significant amount of user feedback to draw statistically significant conclusions. Secondly, there is a risk of negatively

impacting the user search experience if the model performs poorly. Consequently, offline evaluation is commonly used to select candidates for online experiments.

► **中文譯文** 評估搜尋引擎結果的品質已成為一項具有挑戰性的任務，尤其是在判斷哪個排序模型在特定商業指標下表現最好時 [2, 8]。評估模型表現最可靠的方法是線上 A/B 測試，但這種做法有其限制 [5, 7]。第一，由於流量成本高昂，在特定時間內只能比較有限數量的模型，因為要得出統計上顯著的結論，需要大量的使用者回饋。第二，若模型表現不佳，可能對使用者的搜尋體驗造成負面影響。因此，離線評估通常被用來為線上實驗篩選候選模型。

Existing work has proposed a variety of techniques for search engine offline evaluation, including relevance, reliability, timeliness, diversity and fairness [3]. Commonly we evaluate the quality of search by using the evaluation metrics focused on relevance, such as Mean Average Precision (MAP) and normalized Discounted cumulative gain (nDCG). These types of metrics simulate how users interact with the search engine result page (SERP) [4, 14]. Nonetheless, a prevalent drawback of offline evaluation lies in its dependence on historical data, and these metrics may not directly measure the quality of search engine, which can result in imperfect correlations with users' search experiences [1, 6, 10, 16].

► **中文譯文** 既有研究已針對搜尋引擎的離線評估提出多種技術，涵蓋相關性、可靠性、時效性、多樣性與公平性 [3]。我們通常以聚焦於相關性的評估指標來衡量搜尋品質，例如平均精確率均值（MAP）與正規化折扣累積增益（nDCG）。這類指標模擬的是使用者與搜尋結果頁（SERP）互動的方式 [4, 14]。然而，離線評估普遍的缺點在於依賴歷史資料，這些指標未必能直接衡量搜尋引擎的品質，因此可能與使用者實際的搜尋體驗產生不完全吻合的相關性 [1, 6, 10, 16]。

Recent advancements in semantic search and generative AI have significantly transformed the landscape of search systems, particularly with the integration of large language models (LLMs) in ranking and recommendation tasks [9, 15, 17]. LLMs are also explored for relevance judgments and evaluation in information retrieval [12]. The literature has shown that LLMs can closely replicate human judgements through prompt strategies [13] and highlights the growing role of LLMs in automated evaluations [11].

► **中文譯文** 語意搜尋與生成式 AI 近來的進展，已大幅改變了搜尋系統的樣貌，尤其是大型語言模型（LLM）被整合進排序與推薦任務之後 [9, 15, 17]。LLM 也被探索用於資訊檢索中的相關性判斷與評估 [12]。文獻顯示，透過提示詞策略，LLM 能相當貼近地複製人類的判斷結果 [13]，也凸顯 LLM 在自動化評估中日益吃重的角色 [11]。

3 ON-TOPIC RATE | 切題率

In this section, we formulate our task and introduce our proposed metric - On-Topic Rate (OTR). This metric is proposed as the direct measurement to evaluate if the retrieved document is primarily about the query.

► **中文譯文** 在本節中，我們將問題形式化，並介紹我們提出的指標——切題率（On-Topic Rate, OTR）。此指標被提出作為直接衡量「回傳文件是否主要與查詢有關」的方法。

3.1 Task Formulation. Given a member query q as input, the search engine returns a list of documents $D = (d_1, d_2, \dots, d_n)$. Since we focus on content search, here each document corresponds to a post or article on LinkedIn.

► **中文譯文** 3.1 任務形式化 給定會員查詢 q 作為輸入，搜尋引擎回傳一份文件清單 $D = (d_1, d_2, \dots, d_n)$ 。由於我們聚焦於內容搜尋，此處每份文件對應到 LinkedIn 上的一則貼文或一篇文章。

3.2 Computation. On-topic rate is a metric that measures the relevance of search results to the user's query. This metric helps evaluate how well the search engine is able to understand the query's intent and provide relevant results. A high on-topic rate indicates that content search is performing well and providing useful results to our members, while a low on-topic rate suggests that improvements may be needed to better understand the user's search intent. We define OTR for $\langle \text{query}, \text{doc} \rangle$ pair as the following:

$\text{OnTopicRate}(q, d_i) = 1$ if the pair is relevant, otherwise 0 (1)

► **中文譯文** 3.2 計算方式 切題率是衡量搜尋結果與使用者查詢相關程度的指標。此指標有助於評估搜尋引擎理解查詢意圖並提供相關結果的能力。切題率高，代表內容搜尋表現良好、能為會員提供有用的結果；切題率低，則顯示可能需要改善對使用者搜尋意圖的理解。我們將 $\langle \text{query}, \text{doc} \rangle$ 配對的 OTR 定義如下：

切題率(q, d_i) = 若配對相關則為 1，否則為 0（式 1）

3.2.1 OTR@K. We measure the on-topic rate for the ML model by selecting the top K returned documents for each query. We define OTR@K as the total number of query-document pairs that are relevant divided by the total number of returned documents.

$\text{OTR@K} = (\sum_{i=1}^K \text{OTR}(q, d_i)) / K$ (2)

► **中文譯文** 3.2.1 OTR@K 我們藉由選取每個查詢回傳結果中排名前 K 筆的文件，來衡量該 ML 模型的切題率。我們將 OTR@K 定義為：相關的查詢—文件配對總數，除以回傳文件總數。

$\text{OTR@K} = (\sum \text{相關配對數}, i=1 \text{ 到 } K) / K$ （式 2）

4 SEMANTIC EVALUATION SETUP | 語意評估架設

Within this section, we detail the process of establishing semantic evaluation and harnessing Generative AI (GAI) to discern the relevance between queries and documents. Figure 1 has illustrated the overall pipeline: our methodology begins with the creation of a test query set, followed by the formulation of prompts provided to the Large Language Model (LLM) and subsequent calculation of OTR based on the GAI-generated outputs.

► **中文譯文** 在本節中，我們詳述建立語意評估、並運用生成式 AI（GAI）判斷查詢與文件相關性的流程。圖 1 呈現了整體管線：我們的方法始於建立測試查詢集，接著擬定要提供給大型語言模型（LLM）的提示詞，最後根據 GAI 產出的結果計算 OTR。

4.1 Create Query Set. We construct the query set used for evaluation by leveraging different resources to comprehensively cover the relevant topics or areas of interest.

► **中文譯文** 4.1 建立查詢集 我們運用不同來源建構用於評估的查詢集，以全面涵蓋相關的主題或關注領域。

4.1.1 Golden Set. The golden set serves as a stable and uniform standard for assessing and benchmarking queries. It includes the aspirational queries and the top queries from production. We include the following types of query into the golden set:

- Top queries: We incorporate common queries, which can be seen as the most popular and indicative keywords that members are searching for.
- Topical queries: Topical queries encompass search inquiries or questions that pertain to specific subjects or topics. We incorporate these into our golden set because they pose a greater evaluation challenge, often being lengthier and more intricate in their intent. Providing high-quality results for topical queries can enhance our members' ability to access valuable knowledge on LinkedIn.

► **中文譯文** 4.1.1 黃金集 黃金集作為評估與衡量查詢表現的穩定、一致標準，其中包含理想查詢與線上正式環境中的熱門查詢。我們納入以下兩類查詢：

- ．熱門查詢：納入常見查詢，可視為會員搜尋時最熱門、最具代表性的關鍵字。
- ．主題查詢：涵蓋與特定主題或議題相關的搜尋詢問或問句。我們將其納入黃金集，是因為這類查詢的評估挑戰較大，通常語句較長、意圖也較複雜。為主題查詢提供高品質結果，能提升會員在 LinkedIn 上取得有價值知識的能力。

4.1.2 Open Set. The Open Set is a dynamically changing set of queries used for evaluation. The source of the open set includes:

- trending queries and newsy queries from production
- some random queries from production, to add diversity

► **中文譯文** 4.1.2 開放集 開放集是用於評估、會動態變動的查詢集合，來源包括：

- ．線上正式環境中的熱門時事查詢
- ．線上正式環境中的部分隨機查詢，藉此增加多樣性

Table 1: Example golden set queries from different sources.

Top queries: covid-19, resume, microsoft excel, we're hiring, work from home

Topical queries: how to create a personal brand, how to stand out in a competitive job market, how do I negotiate my salary

► **中文譯文** 表 1：來自不同來源的黃金集查詢範例

熱門查詢：covid-19、resume（履歷）、microsoft excel、we're hiring（我們正在招募）、work from home（在家工作）

主題查詢：how to create a personal brand（如何打造個人品牌）、how to stand out in a

competitive job market（如何在競爭激烈的就業市場中脫穎而出）、how do I negotiate my salary（原文誤植 negotiate，應為 negotiate；我該如何談薪水）

Table 2 are some example queries from the golden set and open set used in production for evaluation.

Golden Set: improve workplace communication, remote team best practices, how do I get promoted

Open Set: fed raises rates, leadership first, barbie, women ai study

► **中文譯文** 表 2：線上正式環境評估中所用黃金集與開放集的部分查詢範例

黃金集：improve workplace communication（改善職場溝通）、remote team best practices（遠距團隊最佳實務）、how do I get promoted（我該如何升遷）

開放集：fed raises rates（聯準會升息）、leadership first、barbie（芭比）、women ai study（原文如此）

4.2 Get search results for query set. We gather the following information for each query within the specified set: the top $K = 10$ documents from production, along with post textual information and any mentioned article information (including title and article body, if applicable).

► **中文譯文** 4.2 取得查詢集的搜尋結果 針對指定集合中的每個查詢，我們蒐集以下資訊：線上正式環境中排名前 $K = 10$ 筆文件，連同貼文的文字內容，以及任何被提及文章的相關資訊（包含標題與文章內文，若有的話）。

4.3 Formulate the prompt. We construct the prompt for GPT 3.5 to collect feedback from the LLM. Using the inputs collected from previous steps, we define the prompt to encompass three perspectives:

- (1) the definition of On-Topic rate
- (2) detailed guidance for decision making
- (3) query
- (4) post, including all the text information from the posts, including the commentary and any re-shared posts/articles

► **中文譯文** 4.3 擬定提示詞 我們為 GPT-3.5 建構提示詞，以蒐集來自 LLM 的回饋。運用前述步驟蒐集到的輸入，我們設計的提示詞涵蓋以下要素（原文標題寫「三個面向」，但實際列點為四項，第一項即切題率定義本身）：

- (1) 切題率的定義
- (2) 供決策判斷的詳細指引
- (3) 查詢
- (4) 貼文，包含貼文的所有文字資訊，含留言與任何被轉發的貼文／文章

4.3.1 Metric Definition. We begin by defining a metric and then task LLM with making decisions based on our specific requirements. We keep iterating our prompts to improve the accuracy of

the decision. Based on our experiences from production and our prompting practices, we have learnt that providing very precise definition for the request and adding examples with provided reasons can improve the performance significantly.

► **中文譯文** 4.3.1 指標定義 我們先定義一項指標，再交由 LLM 依我們的特定要求做出判斷。我們持續反覆調整提示詞，以提升判斷的準確度。根據線上實務經驗與提示詞撰寫的實作心得，我們發現：為請求提供非常精確的定義，並附上帶有理由說明的範例，能大幅提升表現。

For example, for our task, we tested two prompts in production for our use case:

- Prompt A: Given the post below, is the post strongly relevant to the query?
- Prompt B: Given the post below, is the post primarily about query or strongly relevant to the query?

Despite the subtle differences between the two prompts, Prompt B outperforms Prompt A significantly. This is because Prompt B specifically directs attention to the main subject of both the query and the post. As a result, it reduces the false positive that might occur due to keyword matching. We have included some examples to show the impacts in Appendix A.

► **中文譯文** 舉例來說，針對我們的任務，我們在線上環境測試了兩種提示詞：

- Prompt A：「給定下方貼文，此貼文是否與查詢高度相關？」
- Prompt B：「給定下方貼文，此貼文主要是在講查詢的主題，或與查詢高度相關？」

儘管兩者用詞差異細微，Prompt B 的表現明顯優於 Prompt A。原因在於 Prompt B 特別把注意力導向查詢與貼文兩者的主要主題，因而降低了因關鍵字比對而產生的偽陽性。我們在附錄 A 提供了幾個範例，以呈現這項差異的影響。

We discover that incorporating well-detailed examples alongside the reasons behind the decision can enhance performance and address various problems identified from observed patterns of failure. With no examples in the prompt the results did not align with human judgment. By summarizing the failure pattern, during our prompt iteration, we determine that incorporating examples with explanations of the judgments can boost the generative AI's capacity for better decision making.

► **中文譯文** 我們發現，在提示詞中加入詳盡的範例並附上判斷背後的理由，能提升表現，並解決從觀察到的失敗型態中發現的各種問題。當提示詞中沒有範例時，結果與人類判斷不一致。透過在提示詞反覆調整過程中彙整失敗型態，我們確認：附上判斷理由說明的範例，能提升生成式 AI 做出更好決策的能力。

4.3.2 Guidance. The guidance offers comprehensive instructions for training LLM to make decisions that align with our expectations. This may involve outlining decision criteria and specific requirements, as well as providing examples of corner cases to illustrate decision-making processes.

Here is an example of defining guidance for OTR:

(1) The on-topic decision should not only consider the keyword match between query and post. It

should reflect the semantic matching between query intention and the post details.

(2) The post information is primarily relevant to the user query.

► **中文譯文** 4.3.2 指引 指引提供完整的說明，訓練 LLM 做出符合我們期望的判斷。這可能包含列出判斷準則與具體要求，以及提供邊界案例（corner cases）的範例，以說明判斷過程。

以下是為 OTR 定義指引的範例：

(1) 切題與否的判斷，不應只看查詢與貼文之間的關鍵字是否吻合，而應反映查詢意圖與貼文內容細節之間的語意匹配。

(2) 貼文資訊主要與使用者查詢相關。

4.4 Compute OTR Metrics. From the GAI output, we have three types of information:

- Binary decision: It directly corresponds to if the retrieved post is primarily on-topic of the query.
- Relevance score: It is the score related to the binary decision. The score is in the range of 0 and 1. The relevance score aims to measure the semantic relevance between query and post. The decision should keep the consistency between the relevance score.
- Decision reason: It explains the reason for the binary decision and relevance score. This field is not used for OTR calculation but it is very important to help us iterate the prompt, since it explains the consideration for the GAI decision.

► **中文譯文** 4.4 計算 OTR 指標 我們從 GAI 的輸出中取得三種資訊：

- ．二元判定：直接對應「回傳貼文是否主要切合查詢主題」。
- ．相關性分數：與二元判定相關聯的分數，範圍介於 0 到 1 之間，用來衡量查詢與貼文之間的語意相關程度；判定結果應與相關性分數保持一致。
- ．判斷理由：說明二元判定與相關性分數背後的理由。此欄位不用於計算 OTR，但對我們調整提示詞非常重要，因為它說明了 GAI 做出該判斷時考量的因素。

Table 3 presents an instance from the pipeline where a post mentioning an article introducing tips for self-promotion was retrieved for the query “promotion tips”. Although this example may be classified as on-topic through keyword matching, it does not accurately reflect the query intention. To address this issue, semantic evaluation enables the identification of the primary topic of the post. Based on the decision reason, it can be concluded that the prompt provided instructions for measuring semantic relevance, and the LLM was able to identify that although the keyword matched the query, the post itself did not address the query intention, resulting in a low relevance score and an off-topic binary decision.

Table 3 (Example output): Query: “promotion tips” | Post: “Here are 13 tips to get you over that mental hurdle. #speakup #tips #leadership” | Binary decision: 0 | Relevance score: 0.4 | Decision Reason: The post is about tips for self-promotion and personal branding, and does not directly address the query of “promotion tips”. The mention of “tips” in the hashtags is not specific enough to make a strong connection to the query.

► **中文譯文** 表 3 呈現管線中的一個案例：查詢「promotion tips」（升遷技巧）回傳了一則提及自我推銷技巧文章的貼文。雖然這個例子若單靠關鍵字比對可能被歸類為切題，但它其實並未準確反映查詢意圖。為了解決這個問題，語意評估能辨識出貼文真正的主要主題。從判斷理由可以看出：提示詞提供了衡量語意相關性的指引，讓 LLM 能夠辨識出——儘管關鍵字與查詢吻合，貼文本身卻沒有回應查詢的意圖，因此得到偏低的相關性分數與「離題」的二元判定。

表 3（輸出範例）：查詢：「promotion tips」（升遷技巧） | 貼文：「這裡有 13 個技巧幫你跨過那個心理障礙。#開口說 #技巧 #領導力」 | 二元判定：0（離題） | 相關性分數：0.4 | 判斷理由：這則貼文談的是自我推銷與個人品牌經營的技巧，並未直接回應「升遷技巧」這個查詢；主題標籤中出現的「tips」不夠具體，不足以與查詢建立強烈的關聯。

Our approach consider (query, document) as on-topic only if the binary decision is 1 and the relevance score exceeds a certain threshold (we use 0.5 in production after analyzing the distribution of relevance score). This ensures that we only consider pairs that the GAI has a strong confidence in, and disregard those that fall below the threshold. By taking this rule, we compute OTR@K and nDCG as the final output.

► **中文譯文** 我們的做法是：唯有當二元判定為 1、且相關性分數超過某個門檻時，才將（查詢，文件）視為切題（我們在分析相關性分數的分布後，於線上正式環境採用 0.5 作為門檻）。如此可確保我們只採計 GAI 有高度信心的配對，並剔除低於門檻的結果。依此規則，我們計算出 OTR@K 與 nDCG 作為最終輸出。

5 EXPERIMENT | 實驗

5.1 Human Evaluation on Generated Output. We employ human evaluation as a metric to ascertain the alignment between decisions made by the proposed semantic evaluation pipeline and expert human annotators, aiming to verify the reliability and consistency of the proposed pipeline in delivering high-quality results.

► **中文譯文** 5.1 對生成結果的人工評估 我們採用人工評估作為指標，藉以確認所提出的語意評估管線所做的判斷，與專家人工標註者之間的一致程度，目的是驗證此管線在產出高品質結果上的可靠性與一致性。

Our human evaluation team comprises 10 colleagues, and all team members possess substantial experience in content search to ensure the credibility of their decisions. To maintain the quality of the annotation process, annotators underwent a rigorous qualification process. This process included familiarizing themselves with annotation guidelines and assessing 20 representative query-response pairs. Following this, we conducted individual assessments of the annotations submitted and organized group discussions to address any confusions or uncertainties regarding the task. Each annotator was given the task of annotating 50 pairs of queries and documents. Both annotators and the GAI are providing the same information: query, post commentary, and re-shared posts/articles.

► **中文譯文** 我們的人工評估團隊由 10 位同事組成，所有團隊成員都在內容搜尋領域具有豐富經驗，以確保其判斷的可信度。為維持標註流程的品質，標註者需經過嚴謹的資格審核流程：包含熟悉標註指引，以及評估 20 組具代表性的查詢—回應配對。之後，我們對送交的標註逐一檢視，並召開小組討論以釐清任務中任何疑惑或不確定之處。每位標註者被指派標註 50 組查詢與文件配對。標註者與 GAI 取得的資訊相同：查詢、貼文留言，以及任何被轉發的貼文／文章。

To gauge the level of agreement among annotators, we collected annotations for a specific set of randomly selected query-response pairs, which yielded a high degree of agreement among the annotators.

Evaluating a query-documents pair requires our annotators to complete a three-step evaluation:

- (1) Evaluate if the query and the post is relevant
- (2) Provide reasons regarding the judgement if it is considered as irrelevant

► **中文譯文** 為衡量標註者之間的一致程度，我們針對一組隨機選取的查詢—回應配對蒐集標註結果，結果顯示標註者之間有高度的一致性。

評估一組查詢—文件配對，原文稱需要標註者完成「三步驟」評估，但實際僅列出兩項（第三步驟原文未明列，推測與後續彙整討論有關）：

- (1) 評估查詢與貼文是否相關
- (2) 若判定為不相關，提供判斷的理由

We utilize the (query, document) data from the annotation result as input for the proposed semantic evaluation process and then compare the output of the pipeline with human evaluation to assess its consistency. By comparing the results, we report 81.72% consistency with the GAI decision.

► **中文譯文** 我們將標註結果中的（查詢, 文件）資料，作為所提出語意評估流程的輸入，再將管線輸出與人工評估結果比較，以評估其一致性。比較結果顯示，與 GAI 判定的一致率為 81.72%。

5.2 Performance on Validation Set. To ensure the quality of our prompt, the team has compiled a validation set comprising various query types and corresponding post pairs, serving as the baseline for assessing the prompt's effectiveness. We select representative queries from production that are frequently searched by members, including:

- company name queries
- title queries, like data engineer, product manager
- skill queries, like finance, customer services, marketing
- newsy queries, like february jobs report
- other top queries, for example: work from home, open to work

► **中文譯文** 5.2 驗證集上的表現 為確保提示詞的品質，團隊彙整了一份涵蓋各種查詢類型與對應貼文配對的驗證集，作為評估提示詞效果的基準。我們從線上正式環境中挑選會員經常搜尋的代表性查詢，包括：

- ．公司名稱查詢
- ．職稱查詢，例如 data engineer（資料工程師）、product manager（產品經理）
- ．技能查詢，例如 finance（財務）、customer services（客服）、marketing（行銷）
- ．時事查詢，例如 february jobs report（二月就業報告）
- ．其他熱門查詢，例如：work from home（在家工作）、open to work（正在求職）

Total our validation set includes 60 queries, each paired with 10 related posts, resulting in 600 query-post pairs. Our team assesses these pairs and makes a binary determination regarding their topical relevance. This judgment is confirmed by at least three team members. We use this validation set as our standard of truth for gauging the effectiveness of our prompts. Furthermore, we utilize this validation set to identify common failure patterns to keep iterating our prompts. The current prompt used in production achieves 94.5% accuracy on the validation set.

► **中文譯文** 驗證集總共包含 60 個查詢，每個查詢各配對 10 則相關貼文，共計 600 組查詢—貼文配對。我們的團隊評估這些配對，並就其主題相關性做出二元判定；每項判斷至少須經三位團隊成員確認。我們以此驗證集作為衡量提示詞成效的判準基準，並用它找出常見的失敗型態，持續調整提示詞。目前線上正式環境所使用的提示詞，在驗證集上達到 94.5% 的準確率。

6 HOW DO WE USE SEMANTIC EVALUATION TO IMPROVE THE PRODUCT? | 如何運用語意評估改善產品

We have adopted the semantic evaluation pipeline as our offline benchmark for experiments in LinkedIn content search. This serves as the foundation for offline evaluation, measuring whether the trained ML model has correctly captured the query intention. The pipeline is designed to monitor the performance of the served content search model, with a weekly update of the open query set and pipeline run to ensure that the calculated OTR falls within the desired range. We also use this tool to evaluate new trained ML models offline, comparing them to the baseline to quickly test for improvements and provided feedback for iteration.

► **中文譯文** 我們已將此語意評估管線採用為 LinkedIn 內容搜尋實驗的離線基準。它是離線評估的基礎，用來衡量已訓練的 ML 模型是否正確掌握了查詢意圖。此管線的設計目的，是監控線上服務中內容搜尋模型的表現：開放查詢集每週更新、管線每週執行，以確保計算出的 OTR 落在預期範圍內。我們也用這項工具離線評估新訓練的 ML 模型，與基準模型比較，以快速測試改善成效並提供迭代所需的回饋。

In addition, we explore the decision reasons to identify growth opportunities. We collect cases identified as off-topic from the pipeline and identify performance gaps that we can improve.

► **中文譯文** 此外，我們也會深入檢視判斷理由，以找出成長的機會。我們蒐集管線判定為離題的案例，藉此找出可以改善的效能落差。

7 CONCLUSION | 結論

In this work, we introduce a semantic evaluation pipeline for search engine offline evaluation. Our proposed metric, on-topic rate, measures the relevance of search results to the user's query. We also outline the construction of prompts and calculation of the OTR, and demonstrate the high consistency between human evaluation and pipeline output. With this semantic evaluation framework, we are able to directly measure the quality of post-search results and gain a better understanding of our search engine's performance.

► **中文譯文** 在本文中，我們提出一套用於搜尋引擎離線評估的語意評估管線。我們所提出的指標「切題率」，衡量的是搜尋結果與使用者查詢之間的相關程度。我們也說明了提示詞的建構方式與 OTR 的計算方法，並展示人工評估與管線輸出之間具有高度一致性。透過這套語意評估框架，我們得以直接衡量貼文搜尋結果的品質，並更深入理解我們搜尋引擎的表現。

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► **中文譯文** （人名不譯，原文照錄）作者感謝以上所列同仁對本研究的支持與貢獻。

REFERENCES | 參考文獻

依委託：只校正書目文字，不譯作者姓名與刊名。原文共 17 筆參考文獻（含 MAP/nDCG、LLM4Eval、SIGIR 相關論文），完整書目請見原文 PDF 第 4 頁 REFERENCES 欄。

APPENDIX A | 附錄 A：Prompt A 與 Prompt B 效能差異範例

During our experiments, we have noticed the importance of formatting the request into very precise guidance that can help GAI understand the request and leverage it for decision making. For example, for two prompts mentioned earlier, given the same information for query and posts, we are seeing Prompt B outperforms Prompt A by significantly reducing the false positives caused by keyword matching.

► **中文譯文** 在實驗過程中，我們注意到：把請求格式化成非常精確的指引，有助於 GAI 理解請求並據以做出判斷，這件事十分重要。舉例來說，針對前述兩種提示詞，在給予相同的查詢與貼文資訊下，我們觀察到 Prompt B 顯著優於 Prompt A，能大幅降低因關鍵字比對而產生的偽陽性。

Table 4 (3 examples, abridged):

(1) Query “react native” — a long post about side projects/AI/work-life balance mentioning “React

Native” only in passing → Prompt A: On Topic | Prompt B: Off Topic.

(2) Query “manager” — a post about “Twin Transformation” sustainability interview, only tangentially about managers → Prompt A: On Topic | Prompt B: Off Topic.

(3) Query “best practices for managing remote teams” — a workshop invite that substantively covers managing large-scale teams → Prompt A: Off Topic | Prompt B: On Topic.

(Full post texts and all 3 rows are in the original PDF, Table 4, p.6.)

► **中文譯文** 表 4（三個案例，摘要）：

(1) 查詢「react native」——貼文是一篇談副業專案／AI／工作生活平衡的長文，只是順帶提到「React Native」 → Prompt A 判定：切題；Prompt B 判定：離題。

(2) 查詢「manager」——貼文談的是「雙軌轉型」永續發展訪談，只是邊緣性地與經理人有關 → Prompt A 判定：切題；Prompt B 判定：離題。

(3) 查詢「best practices for managing remote teams」——貼文是一則工作坊邀請，內容確實深入談大型團隊管理 → Prompt A 判定：離題；Prompt B 判定：切題。

（完整貼文全文與三列表格請見原文 PDF Table 4, p.6。）